



ViewsInsight2.0: Enhancing Video Retrieval for VBS 2025 with an Automatic Query Generator Powered by Large Language Models

Gia-Huy Vuong^{1,2} , Van-Son Ho^{1,2} , Tien-Thanh Nguyen-Dang^{1,2} ,
Xuan-Dang Thai^{1,2} , Minh-Quan Ho-Le^{1,2} , Tu-Khiem Le³ ,
Minh-Khoi Pham³ , Van-Tu Ninh³ , Cathal Gurrin³ ,
and Minh-Triet Tran^{1,2}  

¹ University of Science, VNU -HCM, Ho Chi Minh City, China
tmtriet@fit.hcmus.edu.vn

² Vietnam National University, Ho Chi Minh City, Vietnam

³ Dublin City University, Dublin, Ireland

Abstract. We present ViewsInsight2.0, an advanced iteration of the ViewsInsight system, purpose-built for the Video Browser Showdown (VBS) 2025. ViewsInsight2.0 retains the core strengths of ViewsInsight while introducing significant enhancements to address previous performance limitations. This optimized architecture is designed to deliver exceptional search capabilities tailored for VBS 2025. At its core, ViewsInsight2.0 integrates the CLIP (Contrastive Language-Image Pre-training) model, trained on DFN-5B. Furthermore, we have refined our temporal query mechanism with a more efficient algorithm and a user-friendly interface. In addition, ViewsInsight2.0 utilizes an automatic query generator powered by open-source large language models to efficiently optimize user input queries.

Keywords: Video Retrieval System · Automatic Retrieval System · Interactive Retrieval System · Automatic Retrieval System · AI-based Assistance · Large Language Models

1 Introduction

Known-item search (KIS) in large multimedia collections is a challenging task, and the Video Browser Showdown (VBS) [12] is a research benchmarking challenge that was proposed to encourage the research community to develop optimal solutions for interactive video retrieval in large-scale multimedia databases, in an open and shared environment. This paper presents ViewsInsight2.0, an advanced iteration of ViewsInsight [15], focused on developing an automated video content retrieval system. In summary, this paper makes the following contributions:

1. We leverage a CLIP [10] (Contrastive Language-Image Pre-training) model trained on DFN-5B. Data Filtering Networks (DFNs) [2] are compact systems engineered to automatically sift through large datasets of uncured information. The model was trained on 5 billion images, selected from a total of 43 billion unfiltered image-text pairs, with 12.8 billion sourced from CommonPool-12.8B and 30 billion additional public pairs.
2. We enhance our temporal query mechanism with a more efficient algorithm, alongside a more user-friendly interface for this feature.
3. We propose an automatic query generator that can autonomously generate and extract key information from input queries, minimizing the need for human intervention.

This paper is structured as follows: Sect. 2 reviews existing retrieval systems. Section 3 provides an overview of our system and discusses the extended features of the predecessor system. Finally, Sect. 4 summarizes our findings and discusses future work.

2 Related Work

Recent advances in video retrieval have produced interactive systems that efficiently locate target frames within vast video content. However, many systems struggle with automatic query optimization. To address this, we aim to build a retrieval system using large language models to autonomously refine user queries, ensuring the most relevant images are retrieved with minimal user input.

Concept-based search enhances image retrieval by annotating images with visual markers like detected objects, captions, and OCR-extracted text. These annotations are indexed to match input keywords with relevant images. Concept-based search also leverages NLP to refine results by extracting relevant concepts from text such as captions and tags, as seen in systems like V-FIRST 2.0 [5]. Semantic search advances image and video retrieval by focusing on understanding the meaning and context of visual content. This enables more precise results compared to traditional methods that rely on surface-level features. Modern systems use vision-language pre-trained models like CLIP [11] to boost performance in tasks like zero-shot image-text retrieval, as seen in VideoCLIP [8], diveXplore [13], and the work of Lokoč et al. [7]. These approaches, including the state-of-the-art BLIP [6] model, have significantly improved retrieval outcomes with deeper semantic understanding.

In 2024, we introduced ViewsInsight [15], a video retrieval system with efficient frame locating based on user queries. While successful, users had to manually refine their searches. This experience inspired the development of an automatic retrieval system that processes queries in the background, reducing human intervention.

3 ViewsInsight2.0 – An Enhanced Version of NewsInsight Powered by Large Language Models

ViewsInsight2.0 builds upon our previous work, ViewsInsight [15], by extending its capabilities to include video retrieval. Section 3.1 gives a glimpse of the ViewsInsight2.0 system, then the advancements made in transitioning from ViewsInsight to ViewsInsight2.0 are elaborated in Sects. 3.2, 3.3, and 3.4.

3.1 An Overview of ViewsInsight2.0

The full capabilities of ViewsInsight2.0 are illustrated in Fig. 1, showcasing significant advancements over its predecessor. The main feature of ViewsInsight2.0 is to automatically generate key information such as temporal data, context prompts, metadata, variants from an input query by leveraging opened-source large language models (specifically Meta-Llama3.1 [1], Stanza [9] and T5-base model [14]).

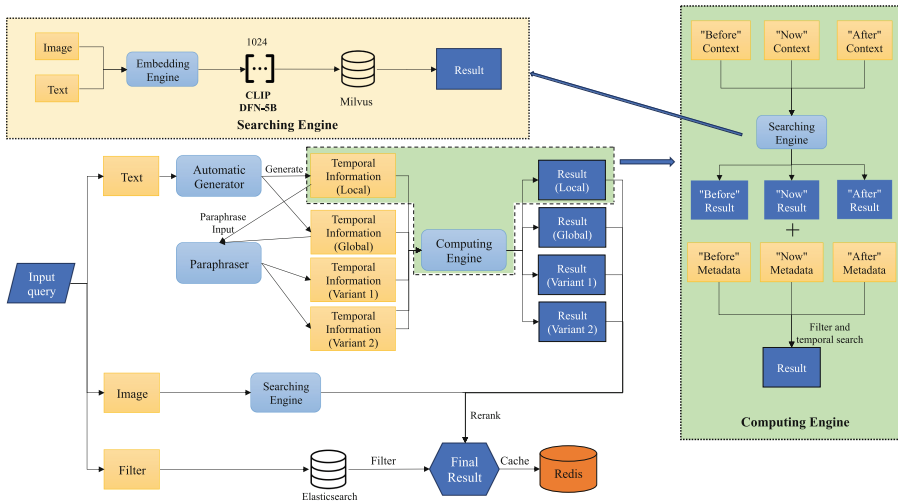


Fig. 1. Overview of ViewsInsight2.0 system.

To illustrate our text processing pipeline with the mentioned key points, we will use the following query in Lifelog Search Challenge [3]: “Greek wine on a Sunday. I was in a Greek restaurant drinking a small bottle of wine and eating Greek food (chips and meat) with a salad. I was sitting at a white tiled table. It was in Thessaloniki in Greece in January 2019.” The query text is first passed through an automatic generator that extracts global and local temporal information. In this case, the context for the current event is identified without before and after contexts. The generator produces the local context “Drinking Greek

wine and eating Greek food at a white tiled table” and the global context “An activity of drinking Greek wine and eating Greek food at a table.”. Besides that, Metadata such as the date (Sunday, January 2019), location (Thessaloniki), and concepts (meat, table, restaurant) are also extracted. These contexts are then passed through a Paraphraser to generate variants such as “A table conversation centered on Greek food and wine” and “Drinks of Greek wine and food at a table.” The global and local temporal information, variants, and metadata are subsequently processed by a Computing Engine to derive specific results. Finally, these results are reranked using an average score across all findings from the previous step to produce the final output. The result of this query displayed on UI is shown in Fig. 2

In summary, ViewsInsight2.0 stands out from its predecessor with two pivotal features: an automatic query generator and a temporal query mechanism. The automatic query generator excels in extracting temporal information, generating context prompts and their variants, and extracting metadata. Meanwhile, the temporal query mechanism utilizes data from each temporal segment generated by the generator to identify the most relevant insights.

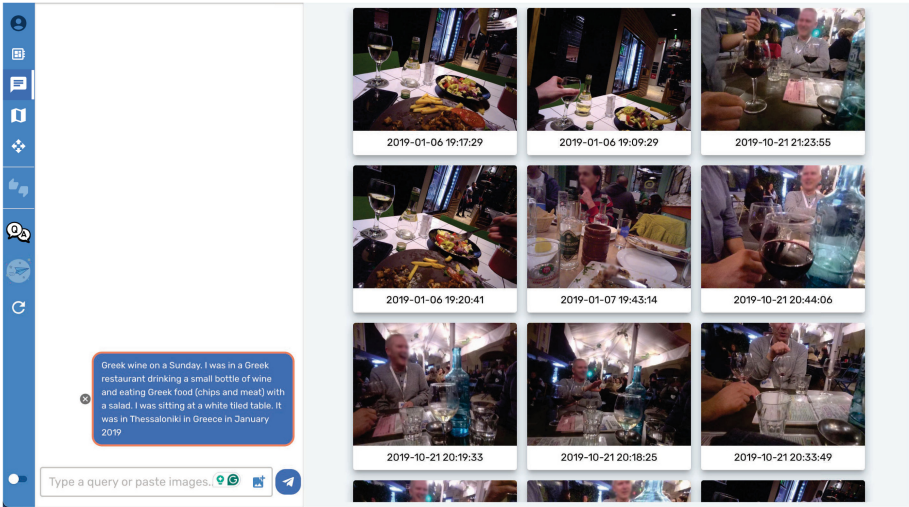


Fig. 2. Illustration of ViewsInsight2.0’s pipeline applied to a query in the Lifelog Search Challenge.

3.2 Challenges from VBS2024

To participate in VBS2025, we enhanced the ViewsInsight system [15] to develop ViewsInsight2.0. Although ViewsInsight excels at retrieving frames from video, it faces three key limitations: slow search performance when using ensemble techniques, the absence of filtering in the temporal query mechanism, and the cumbersome manual data processing required for searches. ViewsInsight2.0 effectively addresses all of these challenges.

3.3 Automatic Query Generator

The Automatic Query Generator in ViewsInsight2.0 integrates three key models: **Llama 3.1** [1], **Stanza** [9], and a **T5-based model** [14]. Each model plays a crucial role in improving the system’s ability to process and understand queries, as illustrated in Fig. 3.

- **Llama 3.1** is tasked with extracting temporal information from input queries into smaller temporal units. For each unit, it generates both local and context-specific prompts, aiding in query contextualization. This not only enables a deeper and more comprehensive understanding of the query’s temporal aspects, but also provides richer input for further stages of processing.
- **Stanza** serves as the system’s metadata extractor. It analyzes the query to identify key elements such as concepts, locations, and time references, providing the system with both the broader context and finer details. Stanza’s robust parsing capabilities allow it to handle diverse input queries, ensuring accurate interpretation across a wide range of user inputs.
- Finally, the **T5-based model** specializes in paraphrasing the context prompts generated by Llama 3.1. By creating alternative phrasing variants, it helps reduce biases during data retrieval and enhances the overall accuracy of the query interpretation process.

3.4 Temporal Query Mechanism

The efficacy of the temporal query mechanism, demonstrated consistently over the years, is particularly evident in the VITRIVR system. Silvan Heller and this team elaborated on this in their paper titled “Multi-Stage Queries and Temporal Scoring in VITRIVR” [4], which underscores the effectiveness of this approach in handling complex queries involving temporal data. Furthermore, this technique’s power is further illustrated in the ViewsInsight system [15], a platform specifically engineered for the Video Browser Showdown 2024 (VBS2024) [12]. ViewsInsight [15] leveraged the temporal query mechanism to achieve superior performance in navigating and retrieving relevant video segments during the competition. In VBS2025, ViewsInsight2.0 improved our temporal query mechanism by introducing event-level filtering, ensuring more accurate results while eliminating redundancies.

The system would require three inputs from the user to use the temporal query mechanism. These include the main query and two temporal descriptions that provide details about events occurring before and after the event in the query. The algorithm then extracts relevant information from these queries within the system. Following this, it updates the scores of the “now” query, taking into account the highest score of the “before” and “after” queries associated with the same video ID. Finally, the algorithm generates a list of search results organized according to their newly updated scores.

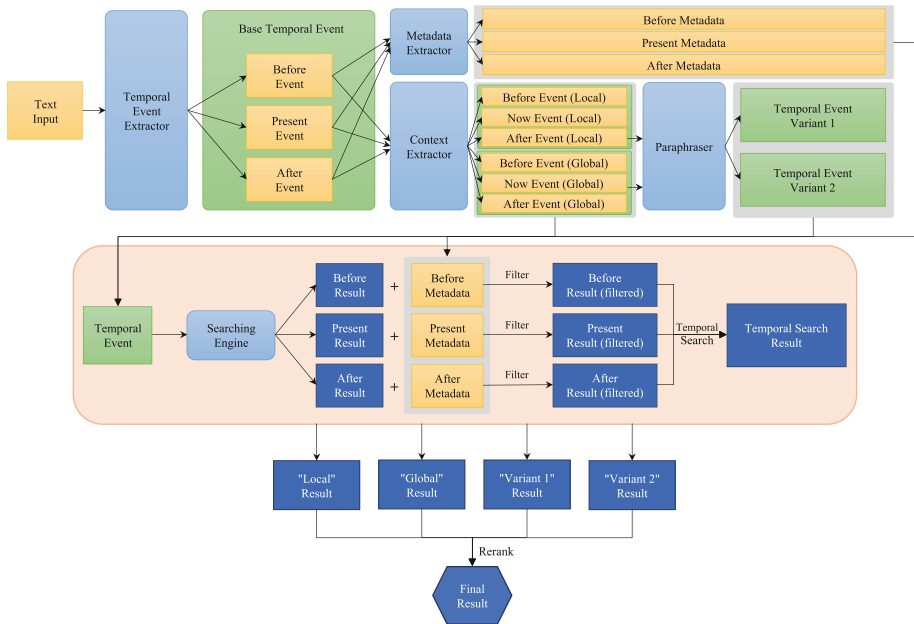


Fig. 3. A breakdown of the Automatic Query Generator for temporal search.

4 Conclusion

In conclusion, ViewsInsight2.0 represents an enhanced iteration of ViewsInsight, specially developed for VBS2025. The system is built around two core innovations: the automatic query generator powered by large language models and a newly introduced temporal query mechanism. The automatic query generator simplifies the user experience by allowing novice users to easily copy-paste queries without the need for manual preprocessing. Meanwhile, the temporal query mechanism strengthens the system’s ability to retrieve the most relevant frames by leveraging the temporal data generated by the query engine. Together, these components enhance both the usability and precision of ViewsInsight2.0 in delivering accurate results.

References

1. Dubey, A., et al.: The llama 3 herd of models (2024). <https://arxiv.org/abs/2407.21783>
2. Fang, A., Jose, A.M., Jain, A., Schmidt, L., Toshev, A., Shankar, V.: Data filtering networks (2023). <https://arxiv.org/abs/2309.17425>
3. Gurrin, C., et al.: Introduction to the seventh annual lifelog search challenge, lsc’24. ACM (6 2024). <https://doi.org/10.1145/3652583.3658891>

4. Heller, S., Sauter, L., Schuldt, H., Rossetto, L.: Multi-stage queries and temporal scoring in vitivr. pp. 1–5 (07 2020). <https://doi.org/10.1109/ICMEW46912.2020.9105954>
5. Hoang-Xuan, N., et al.: V-first 2.0: Video event retrieval with flexible textual-visual intermediary for vbs 2023. In: Dang-Nguyen, D.T., Gurrin, C., Larson, M.A., Smeaton, A.F., Rudinac, S., Dao, M.S., Trattner, C., Chen, P. (eds.) MultiMedia Modeling - 29th International Conference, MMM 2023, Bergen, Norway, January 9–12, 2023, Proceedings, Part I. Lecture Notes in Computer Science, vol. 13833, pp. 652–657. Springer (2023). https://doi.org/10.1007/978-3-031-27077-2_54.
6. Li, J., Li, D., Xiong, C., Hoi, S.: Blip: bootstrapping language-image pre-training for unified vision-language understanding and generation. In: International Conference on Machine Learning, pp. 12888–12900. PMLR (2022)
7. Lokoč, J., Vopálková, Z., Dokoupil, P., Peška, L.: Video search with & nbsp;clip and & nbsp;interactive text query reformulation. In: MultiMedia Modeling: 29th International Conference, MMM 2023, Bergen, Norway, January 9–12, 2023, Proceedings, Part I, pp. 628–633. Springer-Verlag, Berlin, Heidelberg (2023). https://doi.org/10.1007/978-3-031-27077-2_50
8. Nguyen, T.N., et al.: Videoclip: an interactive clip-based video retrieval system at & nbsp;vbs2023. In: MultiMedia Modeling: 29th International Conference, MMM 2023, Bergen, Norway, January 9–12, 2023, Proceedings, Part I, pp. 671–677. Springer-Verlag, Berlin, Heidelberg (2023). https://doi.org/10.1007/978-3-031-27077-2_57
9. Qi, P., Zhang, Y., Zhang, Y., Bolton, J., Manning, C.D.: Stanza: A python natural language processing toolkit for many human languages. CoRR **abs/2003.07082** (2020)
10. Radford, A., et al.: Learning transferable visual models from natural language supervision. CoRR **abs/2103.00020** (2021). <https://arxiv.org/abs/2103.00020>
11. Radford, A., et al.: Learning Transferable Visual Models From Natural Language Supervision (2021)
12. Schoeffmann, K., Lokoc, J., Bailer, W.: 10 years of video browser showdown. In: Chua, T., Wang, J., Tian, Q., Gurrin, C., Jia, J., Zhang, H., Sun, Q. (eds.) MMAsia 2020: ACM Multimedia Asia, Virtual Event / Singapore, 7–9 March, 2021. pp. 73:1–73:3. ACM (2020). <https://doi.org/10.1145/3444685.3450215>
13. Schoeffmann, K., Stefanics, D., Leibetseder, A.: Divexplore at & nbsp;the & nbsp;video browser showdown 2023. In: MultiMedia Modeling: 29th International Conference, MMM 2023, Bergen, Norway, January 9–12, 2023, Proceedings, Part I. p. 684–689. Springer-Verlag, Berlin, Heidelberg (2023). https://doi.org/10.1007/978-3-031-27077-2_59
14. Vladimir Vorobev, M.K.: A paraphrasing model based on ChatGPT paraphrases (2023)
15. Vuong, G.H., et al.: Viewsinsight: enhancing video retrieval for vbs 2024 with a user-friendly interaction mechanism. In: Rudinac, S., et al. (eds.) MultiMedia Modeling, pp. 400–406. Springer Nature Switzerland, Cham (2024)